

Predicting Grade 12 STEM Male Enrollment Using Random Forest Regression

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Summary

This study examined whether the number of Grade 12 male students enrolled in the Science, Technology, Engineering, and Mathematics (STEM) strand could be predicted using school-level characteristics and enrollment-related variables from the Department of Education (DepEd) school dataset. The analysis focused on Senior High School (SHS)-offering schools in the Philippines and applied a Random Forest Regression model to identify important predictors and evaluate predictive performance.

1. Data Selection and Preprocessing

The original dataset contained 60,204 school records and 83 variables. The target variable, `g12_stem_male`, contained approximately 78.7% missing values. Initial inspection indicated that these missing values were largely associated with schools that did not offer Senior High School or did not have applicable Grade 12 STEM enrollment information.

Rather than treating these missing observations as ordinary missing data and imputing them with zeros or statistical estimates, the analysis considered them structural missingness. The dataset was therefore restricted to schools offering Senior High School and observations with valid Grade 12 STEM male enrollment. This resulted in 12,852 observations used for modeling.

This preprocessing step was important because imputing structurally missing values could incorrectly represent schools without applicable STEM enrollment as schools with zero STEM students.

2. Distribution of Grade 12 STEM Male Enrollment

Exploratory analysis revealed a highly uneven distribution of Grade 12 STEM male enrollment.

Approximately 61.97% of the selected SHS-offering schools reported zero Grade 12 STEM male students. This indicates that offering Senior High School does not necessarily mean that a school has male students enrolled in the STEM strand. Schools may instead concentrate their enrollment in other academic or technical-vocational tracks, such as ABM, HUMSS, or TVL.

The target variable was also strongly right-skewed. While the median enrollment was zero, some schools reported substantially larger STEM cohorts, with the maximum reaching approximately 717 male Grade 12 STEM students. This suggests substantial differences in STEM enrollment capacity across schools and creates a challenging prediction problem because a relatively small number of schools account for very large enrollment values.

3. Random Forest Regression Model

The data were divided into an 80% training set and 20% testing set, resulting in 10,281 observations for training and 2,571 observations for testing.

A Random Forest Regression model was selected because it can capture nonlinear relationships and interactions among school characteristics without requiring assumptions of linearity. Feature scaling was also unnecessary because tree-based models are not sensitive to differences in numerical scale.

The model achieved a:

- RMSE: 22.98 students
- MAE: 9.77 students

The difference between the RMSE and MAE reflects the influence of large prediction errors. Because RMSE squares the prediction errors, unusually large errors from schools with very high STEM enrollment have a much greater effect on RMSE than on MAE.

The MAE of approximately 9.77 indicates that the model's predictions were, on average, approximately 10 students away from the actual Grade 12 STEM male enrollment.

4. Prediction Limitations for High-Enrollment Schools

Analysis of the actual-versus-predicted values revealed an important limitation of the Random Forest model. The model performed considerably better for schools with relatively low or moderate STEM enrollment but tended to underpredict schools with extremely large STEM cohorts.

Schools with actual enrollment values between approximately 400 and 700 students frequently received predictions substantially below their actual values.

This behavior is consistent with a fundamental characteristic of tree-based regression models. A Random Forest generates predictions based on averages of observations

within terminal nodes. Consequently, it generally cannot extrapolate far beyond the range represented by the training observations in the relevant terminal nodes.

Therefore, while the model provided useful predictions for the majority of schools, it had difficulty estimating the enrollment of unusually large or highly concentrated STEM institutions.

5. Important Predictors

Variable-importance analysis identified the previous year's Grade 11 STEM enrollment as the strongest source of predictive information.

In particular, `g11_stem_male` and `g11_stem_female` were among the most important predictors of Grade 12 STEM male enrollment. This indicates a strong relationship between the size of the Grade 11 STEM cohort and the subsequent Grade 12 STEM population.

Other enrollment variables, particularly Grade 11 ABM and HUMSS enrollment, also appeared among the important predictors. These variables may provide information about the overall size and academic-track composition of a school.

The importance of `sector_Public` further suggests that school-sector classification contributes to differences in STEM enrollment patterns between public and private institutions.

6. Key Findings

The analysis produced several important findings:

1. Grade 12 STEM male enrollment is highly concentrated.
Most SHS-offering schools have no male Grade 12 STEM students, while a small number of schools account for very large STEM cohorts.
2. Previous Grade 11 STEM enrollment is the strongest predictor.
The size of the previous year's STEM cohort provides substantial information for predicting Grade 12 STEM enrollment, reflecting the importance of year-to-year student progression.
3. School size and track composition provide additional predictive information.
ABM and HUMSS enrollment variables were also important, suggesting that the overall size and academic structure of a school are associated with its STEM enrollment.

4. Public and private schools exhibit structural differences.
The importance of the public-sector indicator suggests that school sector contributes to the observed variation in STEM enrollment.
5. The Random Forest performs well for typical schools but struggles with extreme cases.
The relatively low MAE indicates reasonable predictive accuracy overall, but the model systematically underpredicts

References

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